

TrES-3b

R-1952

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_140419 · created 2026-08-06T15:44:57+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456766.9334 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.056
Transit depth (Rp/Rs)^2	3.83 +/- 2.21 [%]
Orbital Inclination (inc)	84.77 +/- 2.84
Airmass coefficient 1 (a1)	1.63 +/- 0.14
Airmass coefficient 2 (a2)	-0.44 +/- 0.37
Scatter in the residuals of the lightcurve fit is	8.86 %
Best Comparison Star	#2 - [534, 160]
Optimal Aperture	2.31
Optimal Annulus	4.09
Transit Duration (day)	0.071 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.056 vs 0.16309 (deviation 0.59σ)
Duration fit vs published	0.071 d vs 0.0567 d (deviation 0.75σ)
Mid-time O–C	-3.4 min
Depth SNR	3.5
Residual scatter	8.86 %

Light curve

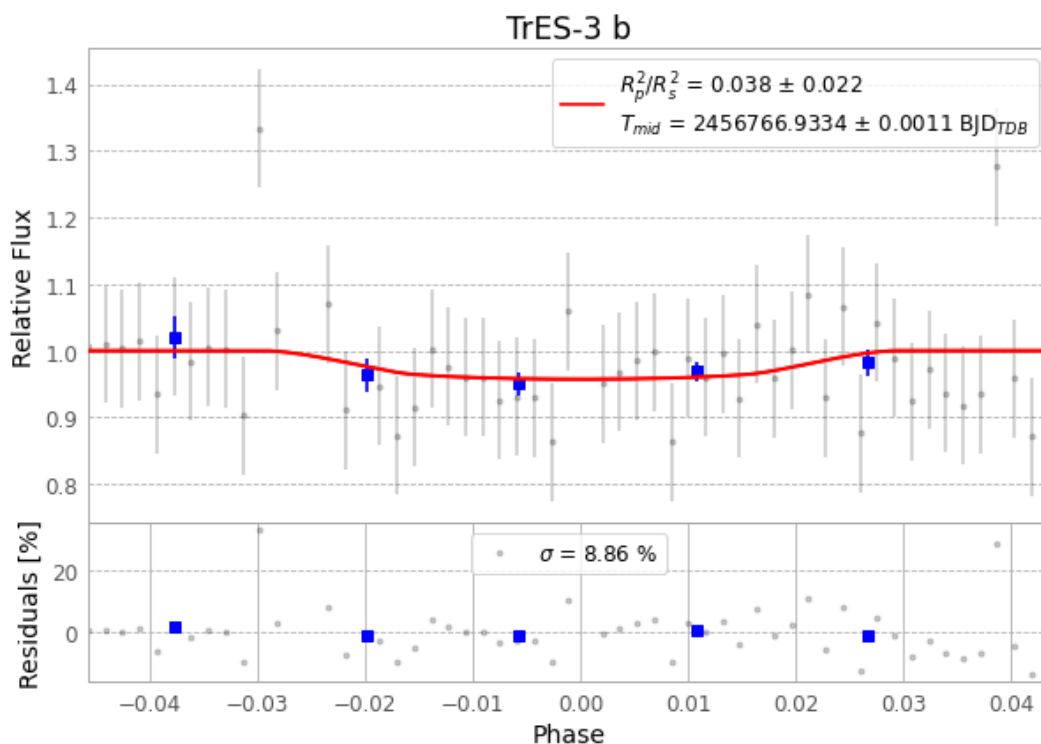


Figure 1 — FinalLightCurve_TrES-3 b_2014-04-19.png (SHA-256 ea7f32bef05728b8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [329, 269], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 aa9519a84e1eec2c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

58 FITS frames (38 MB), TRES-3140419085415.FITS → TRES-3140419114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140419.log (7e7750b04645...), FinalLightCurve_TrES-3 b_2014-04-19.png (ea7f32bef057...), FinalLightCurve_TrES-3 b_2014-04-19.csv (77a04740c05d...)

Compute resources

Wall time 255.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 15:44Z.